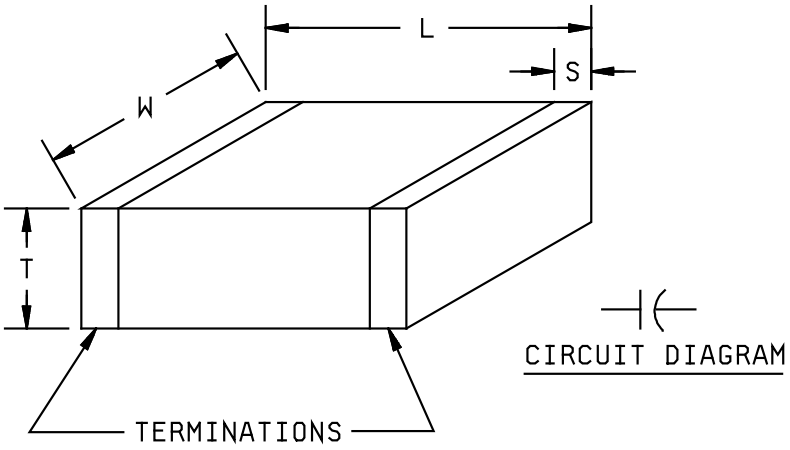
INCH-POUND
MIL-PRF-123/11G
3 June 2026
SUPERSEDING
MIL-PRF-123/11F
w/AMENDMENT 1
5 November 2020

PERFORMANCE SPECIFICATION SHEET

# CAPACITOR, FIXED, CERAMIC DIELECTRIC (TEMPERATURE STABLE AND GENERAL PURPOSE), HIGH RELIABILITY, NONLEADED, STYLE CKS52

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-123.




| L | W | T | | S |
|---|---|---|---|---|
| ± .015 | ± .015 | Min | Max | ± .010 |
| .120 | .100 | .020 | .065 | .020 |

| Inches | mm |
|---|---|
| .010 | 0.25 |
| .015 | 0.38 |
| .020 | 0.51 |
| .065 | 1.65 |
| .100 | 2.54 |
| .120 | 3.05 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are in millimeters and are given for general information only.
3. Dimensions and tolerances are for bare chips. For solder coated terminations, add .025 inch (0.64 mm) to the positive length (L) tolerance, .015 inch to the positive width (W) and thickness (T) tolerances, and .0125 inch (0.32 mm) to the positive S dimension tolerance.

FIGURE 1. Style CKS52 capacitors.

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FSC 5910
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REQUIREMENTS:

Dimensions and configuration: See figure 1.

Case type: Multilayer, unencapsulated, monolithic ceramic.

Product level: A or T in accordance with MIL-PRF-123.

Termination: M, G, S, or Z in accordance with MIL-PRF-123.

Capacitance value: See table I.

Capacitance tolerance: See table I. (F = ± 1 percent; J = ± 5 percent; K = ± 10 percent; M = ± 20 percent.)

Voltage-temperature limit: BP and BX in accordance with MIL-PRF-123.

Operating temperature: -55°C to +125°C.

Rated voltage: See table I (B = 50 Vdc, C = 100 Vdc).

Marking: In accordance with MIL-PRF-123.

TABLE I. Style CKS52 characteristics.

| Part or Identifying Number (PIN) 1/ | Capacitance (pF) | Capacitance tolerance | Voltage-temperature limits | Rated voltage (volts, dc) |
|---|---|---|---|---|
| M123 - 11BP - 301 - - | 300 | F, J, K | BP | 50, 100 |
| M123 - 11BP - 331 - - | 330 | F, J, K | BP | 50, 100 |
| M123 - 11BP - 361 - - | 360 | F, J, K | BP | 50, 100 |
| M123 - 11BP - 391 - - | 390 | F, J, K | BP | 50, 100 |
| M123 - 11BP - 431 - - | 430 | F, J, K | BP | 50, 100 |
| M123 - 11BP - 471 - - | 470 | F, J, K | BP | 50, 100 |
| M123 - 11BP - 511 - - | 510 | F, J, K | BP | 50, 100 |
| M123 - 11BP - 561 - - | 560 | F, J, K | BP | 50, 100 |
| M123 - 11BP - 621 - - | 620 | F, J, K | BP | 50, 100 |
| M123 - 11BP - 681 - - | 680 | F, J, K | BP | 50, 100 |
| M123 - 11BP - 751 - - | 750 | F, J, K | BP | 50, 100 |
| M123 - 11BP - 821 - - | 820 | F, J, K | BP | 50, 100 |
| M123 - 11BP - 911 - - | 910 | F, J, K | BP | 50, 100 |
| M123 - 11BP - 102 - - | 1,000 | F, J, K | BP | 50, 100 |
| M123 - 11BP - 112 - - | 1,100 | F, J, K | BP | 50, 100 |
| M123 - 11BP - 122 - - | 1,200 | F, J, K | BP | 50, 100 |
| M123 - 11BP - 132 - - | 1,300 | F, J, K | BP | 50, 100 |
| M123 - 11BP - 152 - - | 1,500 | F, J, K | BP | 50, 100 |
| M123 - 11BP - 162 - - | 1,600 | F, J, K | BP | 50, 100 |
| M123 - 11BP - 182 - - | 1,800 | F, J, K | BP | 50, 100 |
| M123 - 11BP - 202 - - | 2,000 | F, J, K | BP | 50, 100 |
| M123 - 11BP - 222 - - | 2,200 | F, J, K | BP | 50, 100 |
| M123 - 11BPB242 - - | 2,400 | F, J, K | BP | 50 |
| M123 - 11BPB272 - - | 2,700 | F, J, K | BP | 50 |
| M123 - 11BPB302 - - | 3,000 | F, J, K | BP | 50 |
| M123 - 11BPB332 - - | 3,300 | F, J, K | BP | 50 |
| M123 - 11BX - 562 - - | 5,600 | K, M | BX | 50, 100 |
| M123 - 11BX - 682 - - | 6,800 | K, M | BX | 50, 100 |
| M123 - 11BX - 822 - - | 8,200 | K, M | BX | 50, 100 |

See footnote at end of table.



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TABLE I. Style CKS52 characteristics - Continued.

| PIN 1/ | Capacitance (pF) | Capacitance tolerance | Voltage-temperature limits | Rated voltage (volts) |
|---|---|---|---|---|
| M123 - 11BX - 103 - - | 10,000 | K, M | BX | 50, 100 |
| M123 - 11BX - 123 - - | 12,000 | K, M | BX | 50, 100 |
| M123 - 11BX - 153 - - | 15,000 | K, M | BX | 50, 100 |
| M123 - 11BX - 183 - - | 18,000 | K, M | BX | 50, 100 |
| M123 - 11BX - 223 - - | 22,000 | K, M | BX | 50, 100 |
| M123 - 11BX - 273 - - | 27,000 | K, M | BX | 50, 100 |
| M123 - 11BXB333 - - | 33,000 | K, M | BX | 50 |
| M123 - 11BXB393 - - | 39,000 | K, M | BX | 50 |
| M123 - 11BXB473 - - | 47,000 | K, M | BX | 50 |
| M123 - 11BXB563 - - | 56,000 | K, M | BX | 50 |
| M123 - 11BXB683 - - | 68,000 | K, M | BX | 50 |
| M123 - 11BXB823 - - | 82,000 | K, M | BX | 50 |
| M123 - 11BXB104 - - | 100,000 | K, M | BX | 50 |

1/ The complete PIN will include additional letter(s) to indicate product level, voltage, capacitance tolerance, and termination, as applicable.

Changes from previous issue: The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

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Navy - EC
Air Force - 19
DLA - CC
Other - NA

Preparing activity:
DLA - CC

(Project 5910-2026-022)

Review activity:
Air Force - 85

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